

EEG/GSR/Breathing Data: Quality Control and Preliminary Analysis

Experiment 2026 — a person who stutters (experimental group) & a control participant

1. What was recorded

2 October 2026

Four recordings of the same protocol: 36 scalp EEG electrodes + 2 breathing sensors + galvanic skin response (GSR), 500 Hz.

- **The person who stutters (PWS)** — *one individual* (experimental group), recorded on two different days (sessions 1 and 2 — a natural test-retest).
- **A control-group participant** — the recording was interrupted by a NeoRec crash and exists as two files: part 1 (recovered from the instrument’s raw file) and part 2 (resumed from the speech stage).
- Protocol stages: eyes-open rest (3 min) → eyes-closed rest → hyperventilation (3 min) → eyes-open rest → reading → eyes-open rest → narrative speech (~6 min) → eyes-open rest → eyes-closed rest.

2. Data quality

2.1 Dead electrodes — a systematic hardware problem

Between 11 and 13 of 36 electrodes were **disconnected during every recording**: their signal is a perfectly flat line pinned at the amplifier input rail (± 1.6 mV), with zero variability and no brain content whatsoever (Fig. 2). Critically, they were flat *from the very first sample* — i.e. they were never connected that day, and the impedance check did not catch it. The *same ring* of electrodes (Fpz, FT7/FT8, TP7/TP8, CP3/CPz, FCz, P5/P6, POz, $\pm$ FC3/FC4) is dead in all sessions, including the control subject — so this is hardware (electrode batch, cap rows, or connector slots), not preparation technique. Figure 1 shows the electrode-level picture across the four sessions.

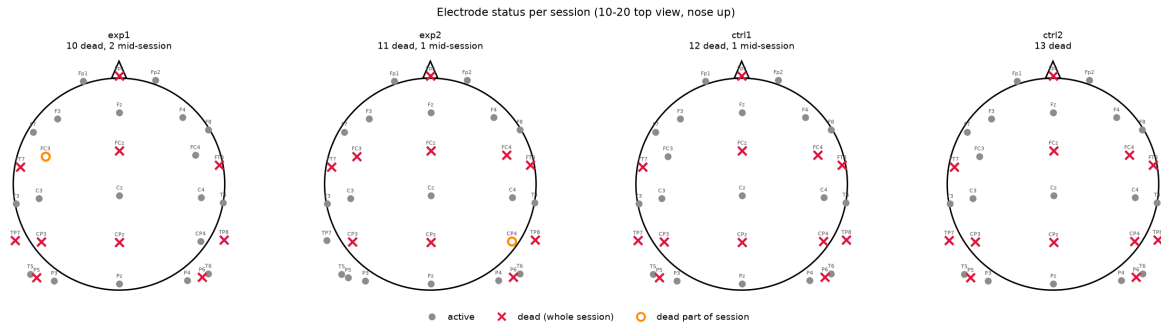


Figure 1: Electrode status per session (10-20 top view). The same ring of electrodes (midline FCz/CPz, lateral FT7/FT8, TP7/TP8, CP3, P5/P6, Fpz) is disconnected in every session and in both subjects; orange marks channels that changed state mid-session (FC3 revived on day 1, CP4 died on day 2, POz flapped in the control part 1).

Mid-session changes (need time-aware bad-channel lists during analysis):

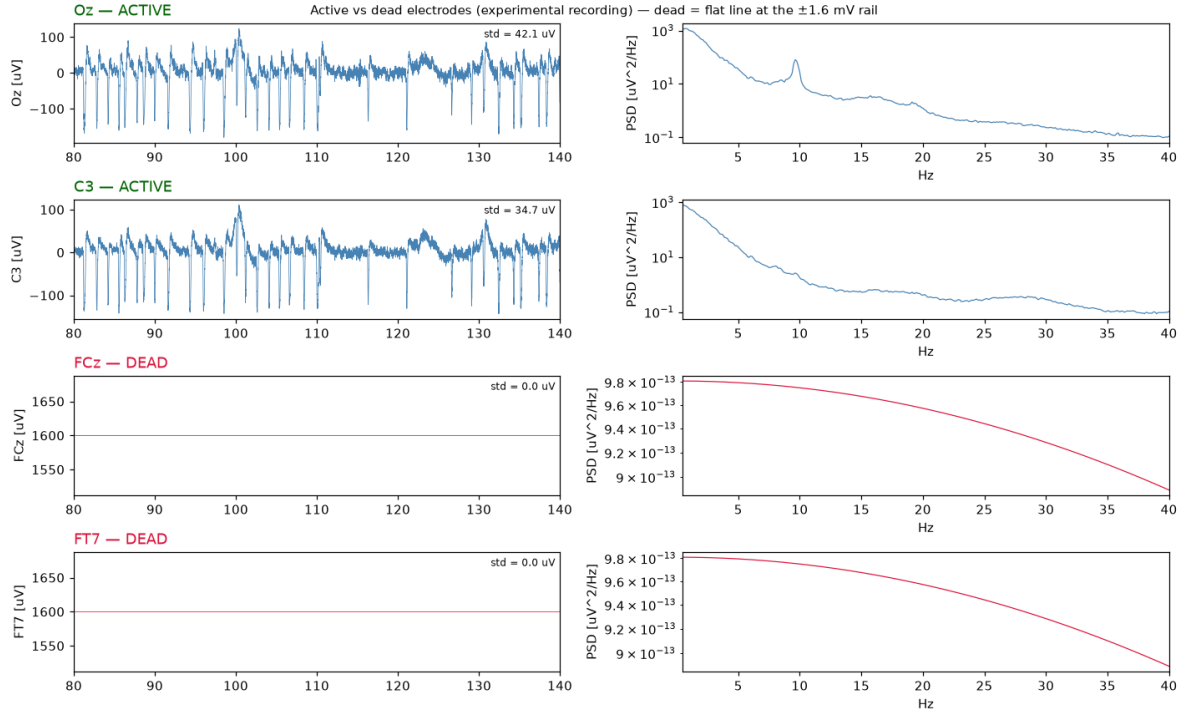


Figure 2: Active vs dead electrodes (the PWS participant, session 1). Active: real brain signal, 1/f spectrum, visible alpha peak. Dead: flat line at the input rail, zero standard deviation, spectrum is numerical noise (16 orders of magnitude weaker).

- PWS s1: FC3 **revived** at ~ 1050 s (during reading); POz dropped at 90 s.
- PWS s2: CP4 died at ~ 1250 s (during speech).
- the control part 1: POz flapping (live 0–140 s, dead 140–1380 s).

2.2 Auxiliary channels

- **Resp2 was not connected in the control session** (confirmed): its trace is pure noise in both parts. The breathing belt (Pesp) is a genuine signal in both parts of the control.
- **GSR of the control part 1 is real** — same amplitude scale and phasic pattern as part 2 (Fig. 3); only the shared crash spike at the end of part 1 made the channels look correlated.
- GSR of PWS session 2 is **very weak** ($\sim 10\times$ below session 1) — phasic responses are visible but the sensor was poorly attached.
- Pesp (1st breathing sensor) is dead in both PWS sessions.

2.3 Stage markers (events)

- PWS s1: all 17 markers present; every fixed-duration stage is 180.0 ± 0.1 s — the marker timing is reliable (tens of ms).
- PWS s2: one marker missing (`open_raw3_start`); the stage is reconstructed from the 179.98 s gap between `reading_finish` and `open_raw3_finish` — exactly one rest block.
- the control part 1: **no markers at all**. Root cause: the raw instrument file contains no event block and no file footer — markers are written when the recording closes normally, and the crash prevented that. The conversion software is not at fault (it preserves markers in normal files). Consequence: restarting the recording from the beginning after the crash would indeed

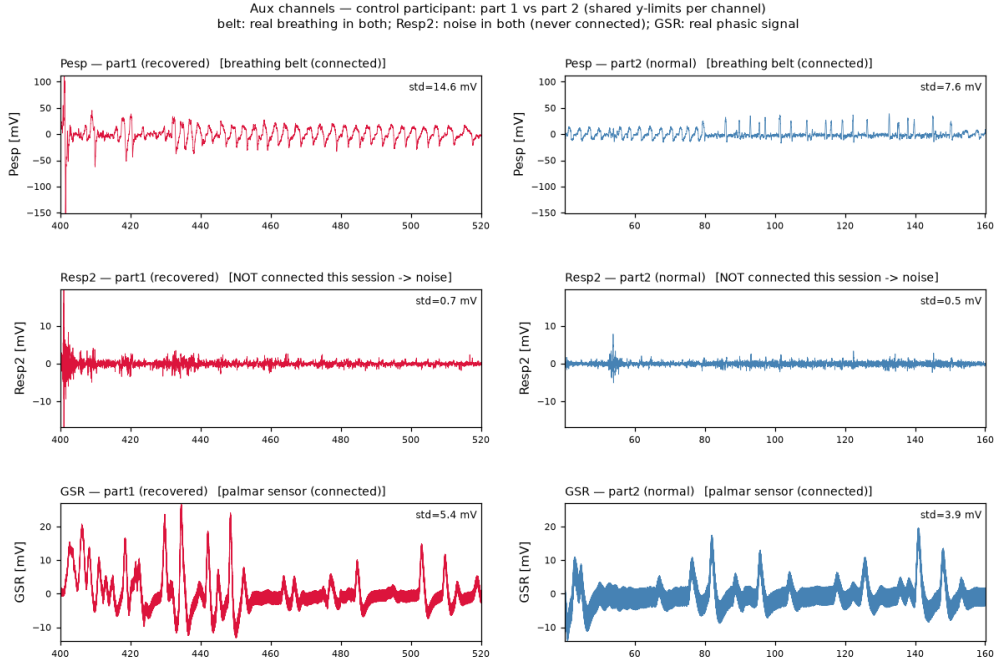


Figure 3: Auxiliary channels of the control participant: part 1 (left) vs part 2 (right), shared scale per channel. Breathing belt: real in both. Resp2: noise in both (never connected). GSR: real phasic signal in both.

have been the right decision; stage boundaries for part 1 were estimated from the data itself (eyes-closed alpha window and the breathing surge match the protocol order) and should be verified against the video.

- “100” markers are redundant end-of-stage signals; all of them land within ≤ 1 s after a named stage end, with identical offsets across sessions — they can serve as backup markers.

3. Does the data look as expected?

- **Alpha rhythm at rest: yes, textbook.** Eyes-closed occipital alpha (8–12 Hz) is 6–20 $\times$ stronger than eyes-open in every recording; a clean 10 Hz peak appears in the spectrum only with eyes closed (Fig. 4). Individual alpha frequency 9.6–9.9 Hz, stable across sessions.
- **Motor artifacts during speech: yes.** High-frequency (30–80 Hz) power is among the highest of all stages in every recording; the recordings are artifact-rich overall (bad contacts move), so artifact rejection must be relative/robust, not by absolute thresholds.
- **Hyperventilation: half-confirmed.** Breathing *depth* doubles-to-triples during hyperventilation (belt amplitude), but *rate* does not increase in any of the three recordings (13–29/min both at rest and during hyperventilation). Either the subjects breathed deep-but-slow, or the belt smooths out fast breathing. Worth checking the video.

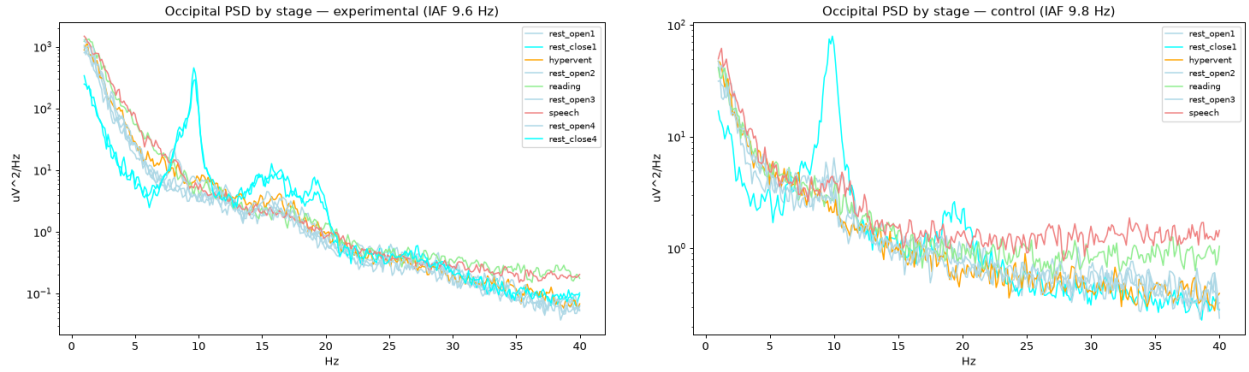


Figure 4: Occipital spectra by stage: eyes-closed curves (cyan) show a dominant 10 Hz alpha peak in both subjects; all other stages lack it.

4. Research-level findings

4.1 Reaction to the hyperventilation challenge

The canonical adult reaction: theta slow-wave “build-up” late in hyperventilation (only about a third of healthy adults show it at all), clearing within 2 minutes after breathing normalizes.

	theta during HV	late HV	3 min after HV	verdict
PWS, day 1	+69%	+98%	still +52%	strong build-up, delayed recovery
PWS, day 2	+2%	+6%	+19%	absent build-up
Control	+29%	+33%	+30%	mild, typical

The *same person* showed opposite reactions on the two days. The most likely explanation is effort/compliance: breathing rate did not rise during hyperventilation in any recording, and the day-2 response looks like half-hearted effort. The day-1 elevation of slow activity lasting more than 3 minutes is the only clinically-flavoured flag; physiologically, CO_2 recovers over ~ 7 minutes, so it may be benign. No high-amplitude rhythmic delta events (the clinically marked kind) occurred. GSR jumped +42% during the challenge (PWS s1) — strong sympathetic arousal. After hyperventilation, eyes-open alpha is elevated in *all three* recordings — a genuine post-challenge state or habituation; a second pre-challenge baseline would separate the two.

4.2 Speech: frontal “effort” theta separates the two people

Frontal-midline theta (4–8 Hz at Fz) is a classic marker of cognitive control and effortful processing. During the narrative speech it rose by **+109% and +69% in the person with stuttering (both sessions)** and by **−4% in the control**. The same pattern appears during reading aloud (+79%/+53% vs +6%). This fits the stuttering literature’s picture of elevated speech-monitoring/effort load, and it replicated across days — the most robust between-subject difference found. Caveat: theta is blink-sensitive; the stuttering subject also blinks much more (see below), so ICA-based cleaning is required before final numbers.

Listening vs answering inside the interview

The interview stage contains both answering and listening to questions. Windows were classified without a microphone by muscle (30–80 Hz) activity: answering windows are the high-muscle bursts, listening windows the quiet stretches between them (45–50% of windows stayed ambiguous and were excluded). The result refines the picture above:

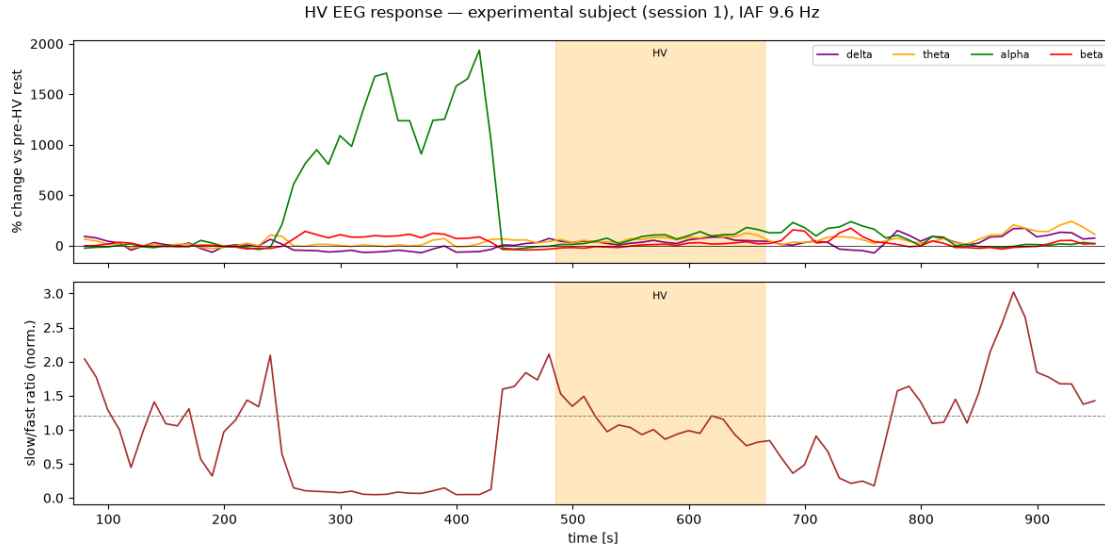


Figure 5: The PWS participant (session 1) around the hyperventilation block. Top: % band-power change vs the pre-challenge eyes-open rest; the large green alpha spike before the shaded zone is the eyes-closed block. Bottom: slow/fast ratio stays elevated for minutes — delayed recovery.

	frontal theta: listening vs rest	answering vs listening	blinks (listen → answer)
PWS s1	+109%	+4%	59 → 59 /min
PWS s2	+71%	+1%	61 → 64 /min
Control	−10%	+13%	16 → 28 /min

In the stuttering subject the frontal-theta elevation is *already maximal while listening* — it is an interview/engagement state, not articulatory effort per se. The control shows the opposite, intuitive pattern: theta near baseline while listening and a modest +13% spike when actually answering. Blinks in the stuttering subject stay high in both states; the control’s blink rate doubles when speaking. This is a single-subject observation with a crude listening/speaking proxy (the interviewer’s voice is also heard during “listening” windows), but it changes the headline: the marker distinguishes *being in dialogue*, not *speaking*.

Language areas (Broca / Wernicke homologues)

With 10-20 scalp electrodes these regions can only be approximated: Broca $\approx$ F7/F3 vs right homologue F8/F4; Wernicke $\approx$ T3/T5 vs T4/T6. Comparing interview windows *against rest* is unreliable here: during the interview *all* bands (alpha included) rise in amplitude — probably sweat lowering electrode impedance — so absolute band changes are not interpretable. The left/right asymmetry *within* the same state is robust to that confound, and it is informative (alpha power reflects relative cortical idling; less alpha = more activation):

alpha L–R asymmetry, %	Broca homologue		Wernicke homologue	
	answer	listen	answer	listen
PWS, day 1	+27	+27	−15	−15
PWS, day 2	+10	+9	−16	−15
Control	−4	+2	+19	+1
Control (resumed)	+2	+2	+9	+9

The stuttering subject shows a consistent pattern on both days: **left Broca homologue relatively “idling”** (positive = more left alpha = less left activation, +10 to +27%) and **right-dominant**

Wernicke homologue (−15%), i.e. a rightward lateralization bias during language — directionally matching the stuttering literature (left frontal/posterior-temporal under-activation with right-hemisphere engagement). The control is balanced or left-leaning. Two caveats: (1) the asymmetry is the same while answering and while listening — again a tonic language-mode state rather than a speech-evoked effect; (2) T3–T6 lie over the temporalis muscle, so the Wernicke numbers are the weaker of the two. Final answers need fMRI-quality localization or at least a high-density cap.

Reading aloud vs dialogue: same effort, different fluency

The subject with stuttering was observed to be nearly fluent while conversing but noticeably disfluent while reading aloud — the opposite of the classic “reading fluency effect” reported in the literature. Comparing the two vocal tasks directly (for the interview, only the speaking windows, so both conditions are vocalizing) gives an informative negative result:

	frontal theta	alpha	blinks	Broca L–R
PWS s1: read → answer	+3%	+10%	46→60/min	+19 → +27%
PWS s2: read → answer	+7%	+20%	46→63/min	0 → +10%
Control: read → answer	−1%	(muscle-inflated)	16→23/min	−4 → −4%

In the stuttering subject the cortical “effort” signature is essentially the *same* in both vocal tasks (frontal theta 14.1 vs 14.5 and 11.2 vs 12.0 μV^2 ; GSR tonic level 278 vs 294 mV, SCR (skin conductance response) rate 13.8 vs 11.0/min) — although fluency clearly differed. So the elevated theta does *not* track stuttering events; it marks the state of vocalizing under observation. The reading-vs-dialogue fluency difference must live at the speech-motor planning level, which scalp EEG with 36 channels cannot resolve — and which no band-power contrast here captures. Disfluency analysis proper requires audio (absent) plus video-annotated stuttering events time-locked to the EEG: the one analysis we would most recommend adding to the protocol next time.

4.3 Arousal: blink rate and GSR

- Blink rate at rest: **30–52/min in the stuttering subject vs 4–16/min in the control** (2–4 $\times$); rises by $\sim 35\%$ during speech in both. Consistent with elevated tonic arousal/anxiety; valid as within-subject comparison.
- Tonic GSR in PWS s1 climbs across the whole session (187 → 294 mV from first rest to speech) and decays only partially after speech — arousal accumulates across the protocol.

4.4 Post-speech “hypoxia-like” effects: none

The 3-minute rest after speech shows no slowing and no alpha decrease relative to pre-speech rest (changes $\leq 1\%$) in any recording. Speech-induced effects here are arousal and effort, and they dissolve within minutes. (The hypoxia-like EEG signature belongs to hyperventilation via CO_2 drop, not to speaking.)

4.5 Other EEG anomalies — screening results

- **No epileptiform-like activity.** An automatic spike/sharp-wave screen plus visual review of the top candidates found only blink artifacts (frontal slow deflections) and muscle bursts (Fig. 6).
- **No focal rhythmic delta episodes** anywhere.
- **No 50 Hz line noise**; no DC/saturation problems beyond the known dead electrodes.

- **One flag:** in the stuttering subject, eyes-closed posterior alpha is markedly right-dominant (parietal P3/P4 ratio -54 to -68% , occipital -23 to -35%), and the pattern *repeats on both days*; the control shows small unstable asymmetries. Could be physiological (alpha asymmetry) or an electrode-contact artifact — check impedance of P3/P4/O1/O2 before interpreting.

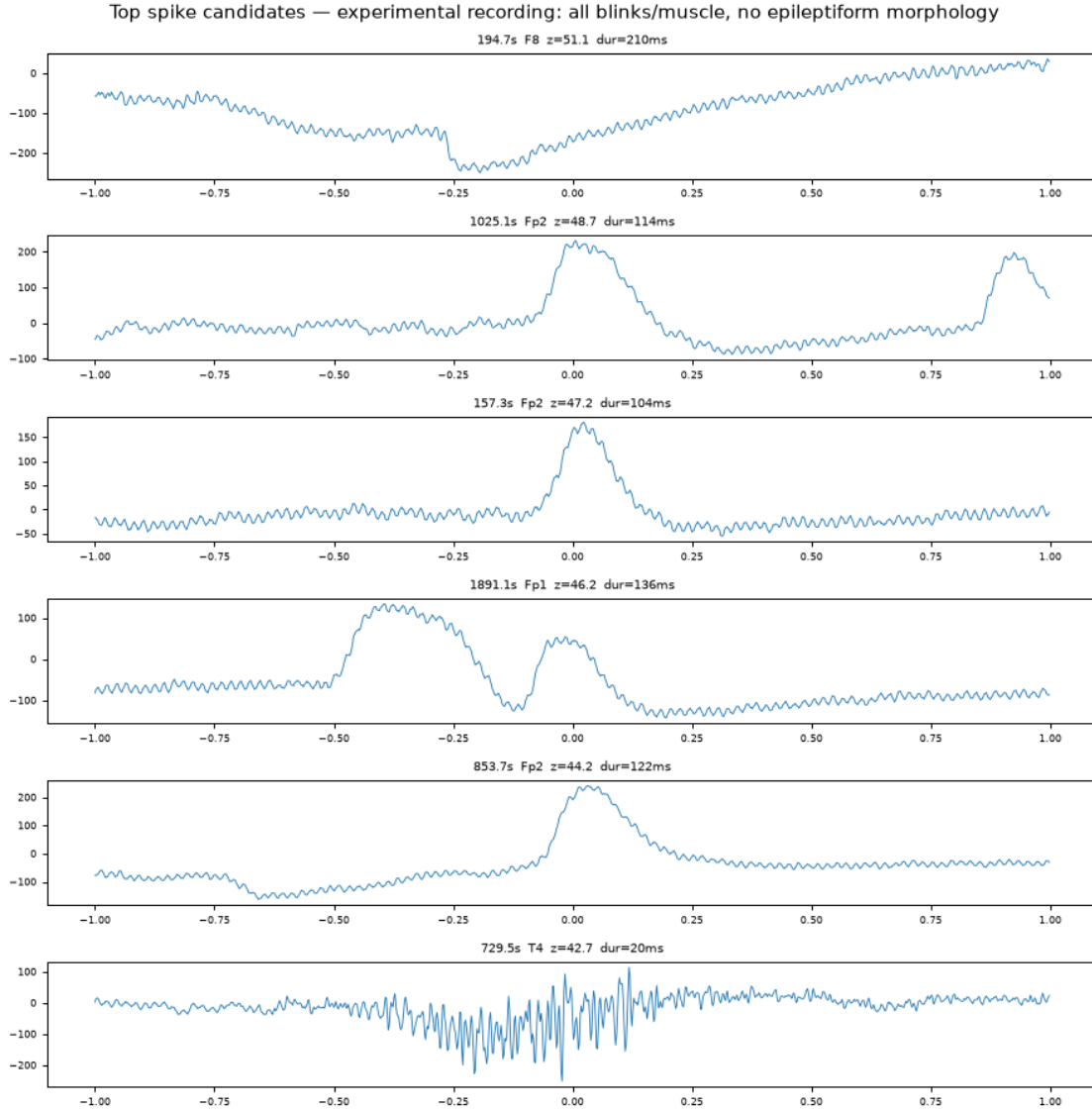


Figure 6: Top automatic “spike” candidates (PWS session 1): all are blinks (Fp1/Fp2, slow positive deflections) or muscle bursts (T4) — no epileptiform morphology.

5. Conclusions and recommendations

1. All four recordings are usable for analysis: PWS s1 fully; PWS s2 with a weak GSR and one reconstructed stage; the control part 1 with estimated stage boundaries (verify vs video) and without Resp2; part 2 fully.
2. Hardware: the dead-electrode ring is identical across sessions and subjects → inspect electrodes/connector slots; find out why the impedance check passes with $\sim 1/3$ of electrodes disconnected.

3. Protocol: verify hyperventilation compliance (video; ideally EtCO₂); add a second pre-challenge baseline; always restart recording after a crash (part 1 markers are unrecoverable).
4. Analysis: time-aware bad-channel lists; robust relative artifact rejection; ICA/CleanLine before quantifying frontal theta and blink metrics.
5. Research leads worth pursuing with more subjects: sustained frontal-theta elevation across the whole interview (present equally while listening and answering) in the stuttering subject, replicated across days, versus a speaking-specific theta spike in the control; elevated resting blink rate; delayed post-hyperventilation recovery; right-dominant posterior alpha asymmetry (after ruling out contacts).